3 **(a)** It is due to the difference in pressure between the bottom surface and top surface. [B1]

(b)(i) $W = mg$

$= V_o \rho_o g$ [B1]

(b)(ii) $U = V_o \rho_f g$ [B1]

(c) In order to sink,

$W > U$ [M1]

$V_o \rho_o g > V_o \rho_f g$

$\rho_o > \rho_f$ [A1]

(d) As the (downward) velocity of the object increases, the (upward) <u>drag force (or fluid resistance) is increasing</u>. [B1]

Hence the downward <u>acceleration (or net force) decreases</u>. [B1]
(also accept: hence the velocity increases at a decreasing rate)

When the (sum of) <u>drag force and upthrust is equal to the weight</u> [B1]
The net force is zero and acceleration is zero, resulting in terminal velocity.